

Profile

Machine Learning Engineer with strengths in graph neural networks and multimodal modeling. Builds reliable training pipelines, rigorous evaluation, and scalable inference. Hands-on with cloud ASR (Azure Speech to Text) and multi-GPU training on Lambda GPU instances. Comfortable owning end-to-end ML workflows—from data to deployed, reproducible results.

Technical Skills

Programming & Data Python, SQL, R, Bash, NumPy, Pandas, feature engineering, data wrangling & augmentation, dataset versioning (DVC), pipeline optimization.

Machine Learning PyTorch, PyTorch Geometric, scikit-learn, Hugging Face Transformers, graph neural networks, attention mechanisms, hyperparameter tuning (Optuna, Ray Tune), model evaluation & validation, ablation studies.

Cloud, Compute & MLOps Azure Cognitive Services (Speech to Text), Distributed Training (multi-GPU & distributed training with PyTorch DDP), Docker, Linux, SSH, Git, GPU optimization (mixed precision, gradient checkpointing), CI/CD, experiment tracking (Weights & Biases), automated inference, cross-validation, performance monitoring, structured logging

Professional Experience

Project Assistant,
DataOceanAI

San Jose, CA
11/2025 - present

- Collaborated with large-scale industry AIML data operations teams to support end-to-end data collection, curation, and delivery for machine learning model development.
- Performed structured data cleaning, normalization, and aggregation using Python and SQL, producing ML-ready datasets aligned with downstream model training requirements.
- Served as a technical liaison between engineering, annotation, and operations teams, translating model-side data requirements into executable data collection and annotation instructions.

Research Assistant,
Illinois Institute of Technology, TIML Research Lab

07/2024 - 12/2025

Supervisor: Prof. Ren Wang & Prof. Can Chen(UNC)

- Designed and maintained modular PyTorch training pipelines for GNN-based bioinformatics models, structuring code into reusable components (data ingestion, model definition, training/evaluation, utilities) to ensure reproducibility and scalability.
- Developed high-throughput data loaders for large, sparse graphs and biological sequences, optimizing preprocessing with vectorized Pandas/NumPy operations, reducing data prep time by 40%+.
- Deployed and orchestrated multi-GPU training workflows on Lambda GPU Cloud, managing distributed data parallelism, environment provisioning (Docker/Conda), sharded dataset handling, and fault-tolerant checkpointing for long-running jobs.

Project Experience

DREAM-GNN: AI-Driven Drug Discovery Platform
Research Project

Research
01/2025 - 07/2025

first author, Paper Published by Briefings in Bioinformatics | [GitHub](#) | [Link](#)

- Built an end-to-end multimodal pipeline for drug-target interaction prediction, combining a BERT-style text encoder with a GNN over molecular/protein graphs in PyTorch Geometric for joint modeling of chemical structures and textual descriptions.
- Implemented cross-attention fusion to integrate SMILES and text embeddings, capturing structural and semantic similarities to improve AUROC and AUPRC over late-fusion baselines.
- Applied augmentation strategies, including random edge drop and feature/sequence masking, to improve generalization in low-label settings and stabilize performance across datasets.
- Achieved 97.38% accuracy ($p < 0.001$) with stratified splits, k-fold CV, and significance testing, validating robustness on multiple benchmarks and consistent convergence across GPUs.
- Optimized input pipelines, batching, and memory allocation for large-scale GPU training under VRAM limits, using mixed-precision and gradient checkpointing to cut memory usage without accuracy loss.

MuSHIN: Advanced Framework for Metabolic Model Reconstruction
Research Project

Research
10/2024 - 07/2025

Co-first author, Paper Accepted in principle by Communications Biology(Nature Profolio) | [GitHub](#) | [bioRxiv](#)

- Designed a reaction-centric hypergraph learning pipeline in PyTorch, building an attention-based HGNN with modular data ingestion, model, and training/evaluation layers to ensure reproducibility, scalability, and easy extension to new datasets.
- Engineered domain-aware feature initialization by integrating ChemBERTa and RXNFP pretrained embeddings into node features before hypergraph propagation, improving convergence speed, reducing overfitting, and boosting downstream accuracy.
- Designed and executed evaluation protocols with k-fold CV, organism-level hold-out splits, and independent dataset tests, improving performance by 17% over baselines and sustaining >90% accuracy under 80% input sparsity.
- Applied model predictions to close metabolic gaps in 24 draft genome-scale models, validating updates through phenotypic forecasts that matched independent fermentation outcomes.
- Built and parallelized a COBRApy-based loopless FBA/pFBA/FVA analysis stack across BiGG and ModelSEED universes with EGC detection, using joblib for multi-core acceleration and cutting evaluation time by >60%.

Education

M.Sc. in Electrical and Computer Engineering,
University of Rochester

Rochester, NY
08/2023 - 05/2025